Data Inspection Report - Hotel Booking Data

1. Dataset Information

- **Dataset Name:** Hotel_Booking_Data
- **Rows (Original):** 141,950
- **Columns:** 32

2. Initial Data Inspection

Inspection Area	Observation
Duplicate Records	Thousands of duplicate rows detected.
Country Code	Country names stored in ISO format.
Null Values	Some column values recorded as NULL
Data Type	A column has a wrong data type

3. Data Quality Issues Identified

i. Null Values

Columns affected

- Agent – 19,555 NULL values
- Company – 133,822 NULL values

ii. Duplicate Records

- 41,202 duplicate rows found.
- Duplicates appeared to be accidental repeat transactions.

iii. Country Code

Country names were stored in ISO 3155–3:2013 format, which can be unique but not descriptive enough at a glance.

iv. Data Type

The **reservation_status_date** had the wrong data type. It should have a **Date** type, but it had a **Number** type.

4. Cleaning Actions Performed

Issue	Action Taken
Null values	Replaced NULL with " ".
Duplicates	Removed duplicate rows.
Incorrect data types	Change data types to appropriate formats

5. Final Dataset Summary

Metric	Before	After
Rows	141,950	100,748
Columns	32	37
Null Values (combined)	153,377	0
Duplicate Rows	41,202	0
Incorrect Data Types	141,950	0

6. Transformations applied:

Step	Cleaning Action Performed	Purpose
1	Combined data from the 2018, 2019, and 2020 tables into a single dataset.	Consolidated all yearly records into one dataset for analysis.
2	Removed duplicate records.	Eliminated duplicate rows to improve data accuracy and prevent double-counting.
3	Created a new **arrival_date** column by merging the **arrival_date_year**, **arrival_date_month**, and **arrival_date_day_of_month** columns.	Combined separate date components into a single complete date field for easier analysis.

4	Changed the data type of **arrival_date** to **Date** and **is_canceled** to **Text**.	Ensured data was stored in appropriate formats for calculations and reporting.
5	Replaced cancellation codes in **is_canceled** (`1` → *Canceled*, `0` → *Not Canceled*).	Improved readability by replacing numeric codes with descriptive text values.
6	Changed the data type of **is_repeated_guest** to **Text**.	Prepared the column for descriptive value replacement.
7	Replaced repeated guest codes (`1` → *Repeat guest*, `0` → *New guest*).	Made the data easier to interpret during analysis.
8	Changed the data type of the **agent** column to **Text**.	Standardized the column before handling missing values.
9	Replaced `"NULL"` values in the **agent** column with blank values.	Removed placeholder text representing missing data.
10	Changed the data type of the **company** column to **Text**.	Standardized the column before cleaning missing values.
11	Replaced `"NULL"` values in the **company** column with blank values.	Cleaned placeholder values to better represent missing information.
12	Changed the **reservation_status_date** column to the **Date** data type.	Enabled date-based calculations and time intelligence features.
13	Created new date-derived columns: **year**, **month_name**, **day**, and **week_of_year** from **reservation_status_date**.	Enhanced the dataset with useful date attributes for trend analysis and reporting.
14	Renamed the generated date columns to **year**, **month_name**, and **day**.	Applied consistent and user-friendly column naming conventions.

7. Observations

The dataset required moderate cleaning due to null values and duplicate records and incorrect data types. All identified issues were resolved using Power Query. The cleaned dataset is now suitable for visualization and analysis in Power BI.